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Research Article

Impacts of arthritis on work performance, functioning and activities of daily living using the Canadian community health survey

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Abstract

Background: Individuals with arthritis often face significant challenges in their activities of daily living and work performance, resulting in substantial effects on both their personal and professional lives.

Objective: This study aims to determine whether workers with arthritis within the Canadian working-age population are more or less likely to miss work, sustain injuries, or experience activity limitations compared to those without arthritis.

Methods: The analysis utilized data from the Canadian Community Health Survey, focusing on employed individuals over 20 from two datasets: 2015–2016 (n = 107,705) and 2017–2018 (n = 112,947). Logistic regression examined occupational health factors, while bivariate analyses via chi-squared tests ($p < 0.01$) filtered variables and Tukey's HSD was used for post hoc analysis ($p < 0.05$). Multinomial models addressed complex responses, and Wald's test compared differences between males and females.

Results: Individuals with arthritis (n = 27,720) were more likely to miss work, experience difficulties in daily functioning and report injury compared to their non-arthritis counterparts. Women with arthritis were also more likely to face difficulties in daily functioning and report joblessness. Men with arthritis were more likely to report work absences but less likely to report injury compared to both women and non-arthritis men.

Conclusion: Despite individuals with arthritis being heavily impacted in terms of functionality, there is a disconnect between accommodations provided and usage. Further research on inclusive accommodations (e.g., ergonomic tools and equipment, flexible work arrangements) and self-management strategies that can account for the problems faced by individuals with arthritis need to be established.

Keywords

arthritis, disease, functional status, work performance, absenteeism, Canada

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Introduction

Arthritis is a common disease that millions of Canadians suffer from daily.[1] Arthritis is characterized by inflammation and joint pain, which can result in body stiffness and difficulty moving.[2] Arthritis describes over 100 arthritic conditions affecting joints and surrounding tissue with osteoarthritis (OA) being the most common, followed by rheumatoid arthritis (RA).[3] Unfortunately, the disease is life-long and currently has no cure. Individuals living with arthritis are faced with day-to-day challenges, including doing housework and running errands, and report needing help with their daily activities.[4]

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While the disease is typically associated with age, a common misconception is associating arthritis primarily with older adults. In fact, roughly 19.8% of the Canadian working population (aged 12 and over) suffer from arthritis with 41.2% of individuals suffering from arthritis in Canada being under the age of 65.[1] With much of the working-age population in Canada encompassing a wide age range (15 to 64 years old), and providing the majority of the country's goods and amenities,[5] arthritis has powerful effects on not just the individual level but for the entire country as well. With the population of individuals aged 18 and over with arthritis increasing by over three hundred thousand people since 2015,[1] prevalence rates of arthritis will continue to be a growing problem for Canada's labor force.[2–4] By 2040, it is projected that approximately 35 million adults (44%) will experience activity limitations attributed to arthritis.[6]

A review conducted by Gulyamova and colleagues found that productivity loss was higher among employees with arthritis, emphasizing the need for further arthritis research for individuals missing work.[7] Alyousef and colleagues found that individuals with arthritis, such as lower limb osteoarthritis, are more likely to experience absenteeism, which negatively affects their work performance and functional capacity.[8] RA is associated with absenteeism-related productivity costs, resulting in $252 million USD lost per year in earnings.[9] Due to the symptoms of the disease posing activity limitations, working-age adults may require lifestyle modifications, devices and/or accommodations for daily activities to compensate for the limitations posed by the disease. Currently, the management of arthritis (e.g., RA) depends on early diagnosis to identify factors to minimize disease progression, but identifying biomarkers to assist early identification still needs some work.[10] While therapeutic approaches using Disease Modifying Anti Rheumatic Drugs (DMARDs) are effective, some individuals may not respond well to the medication.[10] Since the impact of arthritis (e.g., OA pain) is multidimensional, current treatment strategies may not address the full limitations of the disease itself.[11] While current research to address the problems of arthritis is ongoing, it is important to consider what can be done for individuals now. Considering that many individuals have difficulty managing their disease,[12] the current research can benefit from a wider lens on the topic with a focus on functionality, allowing the creation of better management strategies and interventions for individuals arthritis. Specifically, understanding limitations such as reduced mobility, chronic pain, fatigue, and difficulties with manual tasks can inform more targeted approaches that enhance both daily living and work performance. With much of the recent literature lacking a focus on the Canadian working-age arthritis population in terms of functionality as well as gender differences, further investigation is required.

Arthritis has profound effects on an individual and within the context of the workforce and their daily life, function is greatly affected. While pain and injury tend to be clear indicators of function, they are not the only ones to consider, as the disease can have effects on a person life outside of work that can also directly or indirectly affect their work life. Consequently, the recognition of multiple factors and their influence on the impact arthritis has on function, requires a broader perspective. By approaching the topic from a person-environment-occupation (PEO) lens, the complex interactions between the person, environment, and occupational factors that impact an individual's lived experience with arthritis, as well as their occupational performance, are recognized, leading to a wider appreciation of the factors that can affect work and function. A key aspect of this wider appreciation are activities of daily living (ADL), which serve as an indicator of a person's functional status. If impacts upon ADL are present, then the individual may require assistance from others and/or devices.[13] An example of the factors that interact with one another and we seek to investigate can be seen in Figure 1. Using the Canadian Community Health Survey (CCHS), the effects of arthritis on ADLs will be explored in addition to their work life, the prevalence and impact of repetitive strain injuries, and productivity to assess burden. This study aims to assess whether individuals with arthritis are likely to experience work absences, injuries, activity limitations, and difficulties in daily functioning compared to those without it.

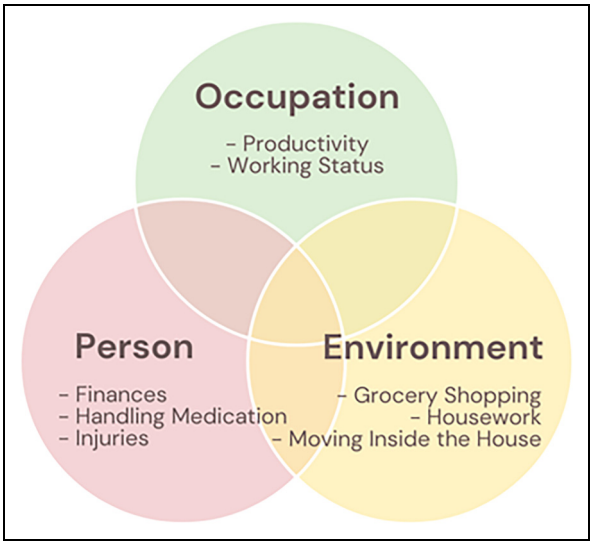


Figure 1. A proposed person-environment-occupation model. The model recognizes that aspects of a person (e.g., handling finances, medication, and having injuries), environmental factors (e.g., navigating grocery shopping, housework, and moving inside the house), and occupational factors (e.g., productivity and working status) can interact to influence occupational performance and each other in terms of function in individuals with arthritis.

Materials and methods

Survey data collection

The CCHS is a cross-sectional self-reported survey that gathers data on health care utilization, health determinants and the health status of the Canadian population.[14] This survey is conducted biennially for health surveillance and research purposes. A multi-stage sample strategy is used to distribute the survey in the ten provinces and three territories within Canada. The multi-stage sample allocation strategy involves provinces being given a power allocation of 0.75 according to the size of their respective population, each province sample is then given a power allocation of 0.35, based on the size of their health region (HR) level. The CCHS sample is then selected using an area frame designed to serve the Labour Force Survey (LFS), a 2-stage stratified design in which each stratum is formed of clusters. The LFS clusters are selected using a sampling method with a probability proportional to size (PPS), and then the final sample is chosen using a systematic sampling of dwellings in the cluster. The CCHS groups these clusters in each HR, leading to a sample of clusters and systematic dwellings being selected in each HR. The survey excludes individuals living on Indigenous reserves and other settlements, full-time members of the Canadian Forces, individuals that are institutionalized, children aged 12–17 years old living in foster care, and individuals living in the Quebec health regions of Région du Nunavik and Région des Terres-Cries-de-la-Baie-James. In total, the excluded population represents less than 3% of the Canadian population aged 12 and over.[14]

Secondary data analysis of the CCHS (2015–2016 & 2017–2018) public use microdata file (PUMF)[14] was performed, giving us a pre-selected cohort, the truncated file largely incorporated the weights within the data with an analytic cohort selected using CCHS data. Survey weights were assigned to each person within the entire sample, with weights corresponding to the number of households in the entire population represented by the respondent.[15] While the CCHS data is largely weighted and validated to provide reliable estimates regarding Canadians, the datasets are limited by their self-reported nature and different datasets may represent different individuals responding to the survey at different points in time. Therefore, we chose to use two datasets (i.e., the 2015–2016 and 2017–2018 datasets) separately to capture a broader range of respondents and more reliable estimates. The survey sample is composed of 107,705 and 112,947 respondents for the 2015–2016 and 2017–2018 datasets, respectively (Tables 1 and 2). The total ages for individuals with arthritis are reported in Table 3.

Variables

The study population included individuals aged 20 years and older who reported being employed at the time of the survey. The present study seeks to investigate whether individuals with arthritis are more or less likely to miss work, sustain injuries, face activity limitations than those without arthritis. Therefore, the variables as shown in Tables 4 and 5 were chosen to measure several aspects of daily experiences that individuals with arthritis may be affected by. These include the reporting of ADLs, such as the difficulty and need for help to complete these ADLs, as well as the inability to do so. Variables within Table 4 measure individuals report experience with such variables. ADLs described in the dataset include difficulty preparing meals, running errands such as shopping for groceries, everyday housework, personal care/taking medication, moving about inside the house, and looking after personal finances. Injuries include repetitive strain injuries (i.e., disorders resulting from repetitive movements, awkward postures, sustained force etc.[16]), limiting normal activities during a 12-month period, a repetitive strain injury while working at a job/business, and other injuries limiting daily activities during a 12-month period. Missed work and productivity includes working at a job/business in the last 3 months, not working at a job/business in the last 3 months due to a specified reason, missing work due to a chronic condition in the last 3 months, missing work due to a chronic condition for a number of days in the last 3 months, and experiencing a loss in productivity.

Statistical analysis

Across the years, the CCHS may decide to replace or remove variables to maintain a reliable and comprehensive source of health data.[14] As a result, data for the variable of productivity and missing work due to a chronic condition in the last 3 months were not measured in the 2017–2018 dataset. Productivity and missing work are important in terms of function in the workplace within our dataset, choosing to drop them without justification would lead to a loss of valuable information and potentially bias the 2017–2018 dataset. To address this, a composite variable was created based on the Pearson's correlation test using data from the 2015–2016 dataset by combining variables with strong correlations to the missing variables ($r > 0.9$). A very strong correlation was found between a decrease in productivity and both repetitive strain injuries in the last 12 months ($r = 0.93$) and other injuries limiting normal activities ($r = 0.93$). As a result, a new composite variable incorporating both factors was created to represent the loss in productivity variable in the 2017–2018 dataset. To ensure consistency across datasets, the new composite variable was also applied to the 2015–2016 dataset. The variables that looked at the reason for why an individual did not work at a job/business for a specified reason within the last 3 months and why an individual missed work due to a chronic condition within the last 3 months had no correlations of a very high strength ($r < 0.8$). Since

Table 1. Workplace demographics of CCHS participants from 2015–2016 (n = 107,705).

	12 to 19		20 to 29		30 to 39		40 to 49		50 to 59		60 to 69		70 to 79		Total	
Ages	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA
Trades	–	243	44	1302	107	1497	236	1346	476	1360	394	681	47	76	1304	6505
Agriculture	–	234	13	589	57	705	125	660	232	718	155	279	14	27	596	3212
Other	–	492	126	3887	396	7054	930	6265	1812	5462	1356	2300	218	308	4838	25,768
Employed	–	4244	308	9838	834	12,451	1812	10,786	3771	10,575	3308	5323	518	736	27,854	79,851

Notes: A = Arthritis, NA = No Arthritis. Data displayed within the table is self-reported. Statistics Canada (2023).

Table 2. Workplace demographics of CCHS participants from 2017–2018 (n = 112,947).

	12 to 19		20 to 29		30 to 39		40 to 49		50 to 59		60 to 69		70 to 79		Total	
Ages	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA	A	NA
Trades	–	211	38	1342	92	1631	213	1391	504	1522	403	817	67	105	1317	7019
Agriculture	–	226	12	600	55	769	101	691	236	750	187	318	27	26	618	3380
Other	–	514	100	4208	353	7576	828	6554	1673	5694	1338	2642	221	359	4513	27,547
Employed	–	3960	248	10,270	695	13,181	1552	11,132	3469	10,891	3263	6004	581	847	9808	56,285

Notes: A = Arthritis, NA = No Arthritis. Data displayed within the table is self-reported. Statistics Canada (2023).

Table 3. Demographics table of individuals With or without arthritis.

	2015–2016		2017–2018	
	WA	WOA	WA	WOA
12 to 19	0	9517	0	10,468
20 to 29	366	11,351	304	11,752
30 to 39	1055	14,077	870	14,854
40 to 49	2246	12,030	1934	12,368
50 to 59	5412	12,660	4910	13,063
60 to 69	8420	11,018	8457	12,386
70 to 79	6476	6222	6951	7120
Male	10,785	39,071	10,620	41,624
Female	16,966	43,737	16,966	43,737

Notes: WA = With Arthritis, WOA = Without Arthritis.

no suitable replacements were found, both variables were excluded from analysis within the 2017–2018 dataset to avoid inaccuracies or inconsistencies into the analysis. Creation of the composite variable for a decrease in productivity provides useful estimations for missing data; however, the variable may suffer from variability in individual reasons and other confounds that could influence its effectiveness.

Two separate methods were used for analysis: logistic regression for variables with binary outcomes (e.g., the survey question/variable gave one or two options to answer) and bivariate analysis via chi-squared tests. The chi-squared tests were used to examine relationships using a conventional threshold of $p < 0.05$ and $p < 0.001$, to balance statistical significance with the risk of Type I errors (false positives). Two logistic regression models were developed for each dataset to explore occupational health factors (i.e., missed work, productivity, activities of daily living and injuries) associated with arthritis in Canada: (1) 2015–2016 dataset and (2) 2017–2018 dataset. A significance level of $p < 0.05$ was used in each of the final models, with post-hoc analysis conducted to adjust the p-values for multiple comparisons. The analysis was completed in RStudio[17] using the 'logistf' package to develop the logistic regression models. Post hoc analysis on significant variables was performed using Tukey's HSD ($p < 0.05$).

For variables that included more than two levels (i.e., the survey question/variable offered three or more different response options), a multinomial logistic regression model was developed to encapsulate the complexity of the outcome. Several multinomial models were developed to examine the relationships between arthritis (predictor) and the aforementioned occupational health factors (outcome). Results were reported as the log-odds ratio and were further exponentiated to give the odds ratio (OR). The z-values were calculated by dividing the estimated coefficients of each multinomial model by their standard error, resulting in a z-value that represents the number of standard deviations each estimated coefficient is from the null hypothesis (indicating no effect). By utilizing the 'pnorm' function in R and the z-values, the cumulative distribution function of the standard normal distribution was calculated. The calculation used the absolute values of the z-values and applied as the formula: 2*(1-pnorm(abs(z_values))). To further account for Type 1 errors, corrections for multiple comparisons were conducted using the False Discovery Rate (FDR) using the 'p.adjust' function in R.

To investigate sex differences between variables, the same methodology previously mentioned was incorporated with the exception of filtering for males or females only to obtain the OR for males and females with arthritis. The OR and their respective standard errors (SE) for both males and females for each variable were logarithmically transformed for further comparison purposes. After transforming the odds ratios into their associated log-odds, pairwise comparisons using the Wald's test were conducted by dividing the difference in log-odds and their respective standard error of difference. The formula for calculation can be seen as: (log(OR_males)) – (log(OR_females)) / sqrt(SE_males^2 + SE_females^2). By utilizing the 'pnorm' function in R and the resulting Wald statistic, the cumulative distribution function of the standard normal distribution was calculated. Corrections for multiple comparisons were applied to the respective p-values by utilizing the 'p.adjust' function in R to apply the FDR method to further control for Type 1 errors, providing a more liberal estimate compared to the Bonferroni correction. The calculation for the p-values used the absolute values of the resulting Wald's statistic and can be expressed as: (p_values <- 2 × (1 –

pnorm(abs(Walds$walds_test)))) and p_values_FDR <- p.adjust (Walds$p_value, method = "fdr") respectively.

Results

Activities of daily living (ADL) (2015–2016)

The analyses revealed significant differences between individuals with and without arthritis (Table 4) in terms of ADL. Arthritic individuals were significantly more likely to report the inability to run errands (OR = 2.88, SE = 0.12, 95% CI [2.27, 3.66]), everyday housework (OR = 4.40, SE = 0.20, 95% CI [3.48, 5.57]), and personal finances (OR = 1.82, SE = 0.13, 95% CI [1.42, 2.32]) compared to individuals without arthritis. Arthritic individuals were also significantly more likely to report difficulty with all ADLs regardless of requiring help or not to complete the task(s) in comparison to their non-arthritis counterparts.

Missing work and productivity (2015–2016)

The analyses revealed significant differences between individuals with and without arthritis (Table 5) in terms of missing work and productivity. Arthritic individuals were significantly more likely to report missing work in the last 3 months than individuals without arthritis (OR = 2.47, SE = 0.100, 95% CI [2.02, 3.01]). Additionally, individuals with arthritis were significantly more likely to report absence from work (OR = 1.36, SE = 0.04, 95% CI [1.27, 1.47]) and not having a job in the last week (OR = 2.75, SE = 0.02, 95% CI [2.66, 2.84]) compared to individuals without arthritis. Individuals with arthritis were found to be significantly more likely to report losses in productivity (OR = 1.20, SE = 0.04, 95% CI [1.12, 1.28]) compared to individuals without arthritis (Table 5).

Injuries (2015–2016)

The analyses revealed significant differences between individuals with and without arthritis (Table 5) in terms of injuries. Arthritic individuals were found to significantly be more likely to report repetitive strain injuries (OR = 1.83, SE = 0.07, 95% CI [1.58, 2.12]) compared to individuals without arthritis.

Activities of daily living (2017–2018)

The analyses revealed significant differences between individuals with and without arthritis (Table 4) in terms of ADL. Arthritic individuals were significantly more likely to report the inability to run errands for all ADLs. Arthritic individuals were also significantly more likely to report difficulty with all ADLs regardless of requiring help or not to complete the task(s) in comparison to their non-arthritis counterparts.

Missing work and productivity (2017–2018)

The analyses revealed significant differences between individuals with and without arthritis (Table 5) in terms of missing work and productivity. Arthritic individuals were significantly more likely to report absence from work (OR = 1.45, SE = 0.04, 95% CI [1.35, 1.56]) and not having a job in the last week (OR = 3.02, SE = 0.02, 95% CI [2.92, 3.12]) compared to individuals without arthritis. Individuals with arthritis were found to be significantly more likely to report losses in productivity (OR = 1.26, SE = 0.02, 95% CI [1.47, 1.59]) compared to individuals without arthritis.

Injuries (2017–2018)

The analyses revealed significant differences between individuals with and without arthritis (Table 5) in terms of injuries. Arthritic individuals were found to significantly be more likely to report repetitive strain injuries (OR = 1.52, SE = 0.02, 95% CI [1.47, 1.59]).

Sex differences (2015–2016)

The analyses revealed several significant differences between men and women with arthritis (Table 6). Women with arthritis were significantly more likely to report difficulty with running errands (Walds = −3.53, $p < .001$) and everyday housework (Walds = −3.09, $p < .05$) even with the help of others compared to their male counterparts. Women with arthritis were also found to significantly be more likely to report not having a job the last week (Walds = −3.56, $p < .05$). Conversely, men with arthritis were significantly more likely to report a work absence the last week (Walds = 3.87, $p < .05$).

Sex differences (2017–2018)

The 2017–2018 dataset revealed far more significant differences between men and women with arthritis compared to the 2015–2016 dataset (Table 7). Women with arthritis were significantly more likely to report difficulty or inability to run errands and everyday housework compared to their male counterparts. Women with arthritis were also more likely to report difficulty with moving about inside the house while requiring help (Walds = −3.17, $p < .001$) and difficulty with everyday housework without help (Walds = −2.34, $p < .05$). Men with arthritis, on the other hand, were significantly more likely to report their working status last week as absent (Walds = 3.51, $p < .001$) but were less likely to report not having a job in the last week (Walds = −3.69, $p < .001$) compared to their female counterparts. Men with arthritis were also

Table 4. Results of the logistic regression and multinomial logistic regression models for activities of daily living (ADL).

	2015–2016			2017–2018		
Variable	Odds Ratio	Standard Error	Confidence Interval	Odds Ratio	Standard Error	Confidence Interval
Difficulty running errands (e.g., groceries)						
Yes, no help needed	3.529**	0.089	2.96, 4.21	4.010**	0.075	3.47, 4.64
Yes, help needed	4.299**	0.089	3.61, 5.12	4.611**	0.070	4.02, 5.29
Cannot do it at all	2.877**	0.122	2.27, 3.66	3.845**	0.102	3.15, 4.69
Difficulty with everyday housework						
Yes, no help needed	4.020**	0.081	3.43, 4.72	4.811**	0.063	4.25, 5.46
Yes, help needed	4.624**	0.087	3.90, 5.49	5.416**	0.067	4.75, 6.17
Cannot do it at all	4.405**	0.200	3.48, 5.57	5.109**	0.101	4.19, 6.22
Difficulty with personal care/taking medication						
Yes, no help needed	4.238**	0.140	3.22, 5.57	3.759**	0.101	3.08, 4.58
Yes, help needed	3.306**	0.124	2.59, 4.22	3.686**	0.108	2.98, 4.56
Cannot do it at all	1.412	0.249	0.87, 2.30	2.606**	0.235	1.64, 4.13
Difficulty moving about inside the house						
Yes, no help needed	6.093**	0.094	5.06, 7.33	7.171**	0.077	6.17, 8.33
Yes, help needed	5.172**	0.206	3.45, 7.75	4.212**	0.179	2.97, 5.98
Cannot do it at all	0.941	0.373	0.45, 1.95	4.096**	0.314	2.22, 7.57
Difficulty looking after personal finances						
Yes, no help needed	2.809**	0.195	1.92, 4.12	1.590**	0.131	1.23, 2.06
Yes, help needed	2.004**	0.130	1.55, 2.58	2.316**	0.089	1.95, 2.76
Cannot do it at all	1.816**	0.126	1.42, 2.32	2.299**	0.119	1.82, 2.90

Notes: *p < .05, **p < .001.

Table 5. Results of the logistic regression and multinomial logistic regression models for missing work, productivity, and injuries.

	2015–2016			2017–2018		
Variable	Odds Ratio	Standard Error	Confidence Interval	Odds Ratio	Standard Error	Confidence Interval
Missed work, chronic, 3 mo (Ref: No)						
Yes	2.47**	0.100	2.02, 3.01	–	–	–
Working status, last week (Ref: Worked at a job, last week)						
Absent from work/business	1.364**	0.038	1.27, 1.47	1.451**	0.037	1.35, 1.56
Did not have a job	2.751**	0.017	2.66, 2.84	3.019**	0.017	2.92, 3.12
Loss in Productivity (Ref: No)						
Yes	1.198**	0.035	1.12, 1.28	1.262**	0.016	1.22, 1.30
Repetitive Strain Injury (12 mo) (Ref:No)						
Yes	1.833**	0.073	1.58, 2.12	1.52**	0.02	1.47, 1.59
Other Injuries limiting normal activities (12 mo) (Ref:No)						
Yes	X	X	X	X	X	X

Notes: X = results found were insignificant during bivariate analysis, "–" = results were not conducted due to no suitable replacement found. "Ref: No" is the reference category of individuals having no such experience with the respective variables. *p < .05, **p < .001.

significantly less likely to report injury compared to individuals without arthritis (OR = 0.91, SE = 0.03, 95% CI [0.86, 0.97]) and were less likely to report injury (Walds = −6.05, p < .001) compared to their female counterparts.

Discussion

In this study, we observed if individuals with arthritis were more likely to miss work, face activity limitations and have difficulty functioning throughout their daily lives than those without arthritis, but have a reduced chance of injury. Our results suggest that individuals with arthritis are significantly affected by arthritis by almost all occupational health factors observed.

Activities of daily living (ADL)

Individuals with arthritis suffer from a variety of problems in their daily living due to the condition. The study findings

Table 6. Results of the logistic regression and multinomial logistic regression models for activities of daily living (ADL), and missing work, injuries and productivity (2015–2016).

	Males				Females				Total Diff.
	OR	SE	Lower CI	Upper CI	OR	SE	Lower CI	Upper CI	Wald's
Difficulty running errands (e.g., groceries)									
Yes, difficulty, no help needed	4**	0.15	2.98	5.36	3.18**	0.11	2.56	3.95	1.22
Yes, difficulty, help needed	2.76**	0.15	2.07	3.69	5.34**	0.12	4.26	6.7	−3.53**
Cannot do it at all	2.31**	0.18	1.63	3.26	3.65**	0.17	2.59	5.14	−1.85
Difficulty with everyday housework									
Yes, difficulty, no help needed	3.51**	0.14	2.66	4.64	4.12**	0.1	3.38	5.01	−0.91
Yes, difficulty, help needed	2.95**	0.16	2.16	4.02	5.33**	0.11	4.32	6.57	−3.09*
Cannot do it at all	3.74**	0.17	2.7	5.19	5.33**	0.18	3.79	7.52	−1.47
Difficulty with personal care/taking medication									
Yes, difficulty, no help needed	4.65**	0.22	3.04	7.1	3.93**	0.18	2.74	5.63	0.60
Yes, difficulty, help needed	2.53**	0.17	1.8	3.56	4.7**	0.19	3.25	6.81	−2.41
Cannot do it at all	1.91	0.34	0.99	3.7	1.06	0.37	0.51	2.21	1.17
Difficulty moving about inside the house									
Yes, difficulty, no help needed	5.53**	0.14	4.18	7.31	6.49**	0.13	5.06	8.32	−0.84
Yes, difficulty, help needed	4.44**	0.3	2.48	7.95	6.01**	0.29	3.39	10.66	−0.73
Cannot do it at all	1.46	0.5	0.54	3.89	0.61	0.57	0.2	1.86	1.15
Difficulty looking after personal finances									
Yes, difficulty, no help needed	2.54**	0.27	1.48	4.35	3.29**	0.28	1.88	5.75	−0.65
Yes, difficulty, help needed	1.93**	0.19	1.33	2.81	2.12**	0.18	1.5	3.01	−0.36
Cannot do it at all	1.45*	0.18	1.03	2.05	2.59**	0.19	1.79	3.76	−2.23
Missed work, chronic, 3 mo									
Yes	3.06**	0.15	2.26	4.1	2.04**	0.13	1.58	2.65	2.02
Working status, last week									
Absent from work / business – last week	1.6**	0.06	1.43	1.81	1.18**	0.05	1.07	1.31	3.87*
Did not have a job – last week	2.52**	0.03	2.39	2.64	2.84**	0.02	2.72	2.96	−3.56*
Loss of Productivity									
Yes	1.18*	0.05	1.06	1.31	1.22**	0.05	1.12	1.34	−0.47
Repetitive Strain Injury (12 mo)									
Yes	1.96**	0.11	1.58	2.43	1.73**	0.1	1.41	2.1	0.85
Other Injuries limiting normal activities (12 mo)									
Yes	X	X	X	X	1.39*	0.11	1.12	1.72	X

Notes: X = results found we insignificant during bivariate analysis. *p < .05, **p < .001. Negative values for the Wald's statistic indicate that the odds ratios for females is higher. Positive values indicate that the odds ratios for males were higher.

show that individuals experience significant difficulty in their ADLs, often relying on help from others for tasks such as running errands like grocery shopping and doing everyday housework (Table 4). Hootman and colleagues found that arthritis has substantial personal effects in correlation with age, 45% of adults suffering from arthritis aged 65 or older face activity limitations related to arthritis with 25% reporting severe pain. 30.6% of working-age adults with arthritis on the other hand report work limitations because of arthritis.[18] Furthermore, individuals with arthritis can experience deterioration in their physical abilities over time. Vermaak and colleagues note that people with arthritis experience a decline in their physical function, possibly leading to no improvements in general health.[19] Naturally, such deterioration can impact an individual's ability to conduct several tasks. While there are methods to reduce the effects of arthritis over time, they will persist into older age. Notably, older adults face ADL challenges in which they typically rely on a family member or caregivers for support in overcoming these difficulties.[20] Future studies should investigate methods to assist individuals with arthritis as they age. These findings are consistent with the results of the study (Table 4).

The study's findings indicate that individuals suffering from arthritis rely on support from others to manage their ADL challenges (Table 4). The lack of mobility and inability to complete tasks independently in tandem with the deterioration of arthritis as a disease (including the reliance on external support) can lead to several psychosocial effects on the individual. One of the biggest being learned helplessness (LH), a psychological state that has patients with a chronic disease experience feelings of helplessness.[21] A study conducted by Berkovic and colleagues found that due to these psychosocial impacts, individuals with arthritis would perceive themselves to be a liability to employers even despite employment, viewing the burden to be huge, at risk of injury and in need of assistance.[12] Furthermore, access to additional accommodations can negatively affect

Table 7. Results of the logistic regression and multinomial logistic regression models for activities of daily living (ADL), and missing work, injuries and productivity (2017–2018).

	Males				Females				Total Diff.
	OR	SE	Lower CI	Upper CI	OR	SE	Lower CI	Upper CI	Wald's
Difficulty running errands (e.g., groceries)									
Yes, difficulty, no help needed	3.27**	0.12	2.56	4.16	4.33**	0.1	3.59	5.22	−1.8
Yes, difficulty, help needed	3.14**	0.13	2.44	4.04	4.99**	0.09	4.22	5.91	−3*
Cannot do it at all	2.83**	0.16	2.05	3.89	4.65**	0.14	3.57	6.07	−2.35*
Difficulty with everyday housework									
Yes, difficulty, no help needed	3.73**	0.11	3.01	4.62	5.12**	0.08	4.38	5.98	−2.34*
Yes, difficulty, help needed	3.49**	0.12	2.74	4.46	5.92**	0.08	5.04	6.94	−3.55**
Cannot do it at all	3.3**	0.15	2.46	4.45	7.65**	0.15	5.73	10.21	−3.98*
Difficulty with personal care/taking medication									
Yes, difficulty, no help needed	3.68**	0.16	2.69	5.03	3.72**	0.13	2.87	4.81	−0.05
Yes, difficulty, help needed	3.02**	0.16	2.22	4.11	4.89**	0.16	3.57	6.68	−2.15
Cannot do it at all	1.48	0.38	0.7	3.14	4.48**	0.34	2.31	8.72	−2.17
Difficulty moving about inside the house									
Yes, difficulty, no help needed	6.48**	0.12	5.15	8.15	7.58**	0.1	6.19	9.28	−1
Yes, difficulty, help needed	2.04*	0.3	1.13	3.67	7.16**	0.26	4.31	11.88	−3.17*
Cannot do it at all	1.96	0.53	0.69	5.58	6.89**	0.45	2.86	16.63	−1.8
Difficulty looking after personal finances									
Yes, difficulty, no help needed	1.35	0.21	0.9	2.04	1.83**	0.17	1.3	2.56	−1.1
Yes, difficulty, help needed	1.94**	0.14	1.48	2.56	2.67**	0.12	2.12	3.37	−1.74
Cannot do it at all	1.91**	0.18	1.35	2.72	2.88**	0.17	2.08	3.99	−1.67
Missed work, chronic, 3 mo									
Yes	–	–	–	–	–	–	–	–	–
Working status, last week									
Absent from work / business – last week	1.67**	0.06	1.49	1.86	1.28**	0.05	1.17	1.41	3.51*
Did not have a job – last week	2.76**	0.03	2.62	2.9	3.13**	0.02	2.99	3.27	−3.69*
Loss of Productivity									
Yes	1.18**	0.02	1.12	1.24	1.35**	0.02	1.29	1.4	−4.06**
Repetitive Strain Injury (12 mo)									
Yes	1.59**	0.03	1.5	1.69	1.48**	0.03	1.4	1.56	1.77
Other Injuries limiting normal activities (12 mo)									
Yes	0.91**	0.03	0.86	0.97	1.17**	0.03	1.11	1.23	−6.05**

Notes: X = results found we insignificant during bivariate analysis, "–" = results were not conducted due to no suitable replacement found. *p < .05, **p < .001. Negative values for the Wald's statistic indicate that the odds ratios for females is higher. Positive values indicate that the odds ratios for males were higher.

social relationships within the workplace.[22] Vergara and colleagues found that LH and self-efficacy (SE) were both related to the degree of disability, pain, and disease activity. However, the relationship between LH and SE with activity of the disease suggests that treatment for arthritis can modify the psychosocial effects individuals can experience.[21] A potential solution to improve self-efficacy in individuals with arthritis is self-management strategies (regulating emotions in a productive manner), such as cognitive-behavioral counselling and exercise therapy.[23] However, for self-management to be effective, individual social circumstances and available resources should be considered.[24] One social circumstance to consider is social support (the access to help and assistance from others that makes a person feel cared for), Holland and Collins note that increased workplace social support can lead to increased job retention and decrease the negative effects on social relationships.[22] In turn, social support can help alleviate the psychosocial impacts that the disease can have upon the individual and allow individuals to be more resistant and resilient to stressful events such as pain.[25] A larger presence of social support within the workplace (e.g., a workplace without the lack of social support) would be required to be investigated further within future research to see if the psychosocial effects (i.e., LH and SE) can be alleviated over time.

Sex differences were found between men and women in terms of ADLs, with women being significantly more likely to report difficulties or the inability to run errands, do everyday housework, and moving about inside the house (Tables 5 and 6). According to a study by Berger, men had higher odds of treatment adherence with specialty medication for treating RA.[26] Perhaps better adherence in men may suggest the increased likelihood for women to be affected in terms of their functionality with ADLs, the symptoms are just more prominent. However, the differences

between men and women for ADLs may be explained by gender norms. Women are more likely to be the primary grocery shopper,[27] do more housework than men,[28] and by extension move around more at home. Especially when considering that women are held in higher regard than men during tasks such as cleaning their rooms,[28] perhaps the increased difficulty stems due to an increased loss in standards and not functionality.

Missing work and loss of productivity

Individuals with arthritis were more likely to report that they spent time off work due to absences or not having a job within a week timespan (Table 5). Xavier and colleagues noted that an increase in the activity of arthritis has a significant effect on the increased difficulty to maintain or seek job opportunities especially with workplace disabilities being present.[29] Individuals with arthritis were also found to be more likely to report injury and missed work compared to individuals without arthritis, suggesting a level of self-awareness about the impacts of their disease (Table 5). As injuries become more frequent, the ability to manage work and be present within the workspace becomes more difficult. Berkovic and colleagues found that individuals with arthritis had to reduce their work hours and that navigating the need for sick leave and missing work is difficult, all in an effort to maintain a work-life balance and continue functioning in the workplace.[12] Additionally, hospitalization due to injury is driven by socio-economic factors, the socioeconomic stress due to reduced work hours (and ultimately pay) can play a role in potential injuries as well.[30] A personal dilemma is presented: is it better to risk injury to avoid financial burdens by working, or are the effects of the disease so strong that it requires time off work? In turn, individuals aren't just dealing with their arthritis symptoms but are also caught in a cycle of stress and psychosocial impacts due to the impact the disease has on their workplace.[12]

As the burden on the workplace and the need for risk management to complete tasks increases,[12] the productivity levels of individuals with arthritis are also significantly affected, as found within the present study (Table 5). Productivity can be affected via arthritis by a variety of physical impairments such as joint pain, stiffness, and fatigue.[31] Workers with rheumatoid arthritis (RA) lose roughly four hours of productive work every two weeks on average, based on a 28.7-hour work week, leading to worse mental health and physical limitations.[32] As such, arthritis has led them to a loss of productivity, and an awareness of being prone to injury creates a state of increased cautiousness in the workplace to avoid injury in even basic office tasks.[12]

Naturally, the link between arthritis and missing time off work is due to the natural effects of the disease itself. Even with a state of increased cautiousness, individuals are still prone to the presence of sickness presenteeism (wanting to work despite illness) and sickness absenteeism (taking time off due to illness). Holland and Collins found that sickness presenteeism in a sample of individuals affected by RA occurred involuntarily due to a lack of accommodating policies related to sickness absence policies[33] and despite individuals reporting flareups they believed their symptoms did not require the use of workplace accommodations.[34] Gignac and colleagues note that the lack of use of workplace accommodations was related to their own perceived need,[34] suggesting that they may be downplaying their disability due to concerns about being perceived as a burden.[21] Future research involving work policies for Canadians and their own personal views would need to be conducted to find the systemic problems that lead to nonuse of policies.

Sex differences were also found in terms of missing work and productivity. Men with arthritis were found to be less likely to report injury compared to their female counterparts, however, they are still affected in terms of their functionality as seen in the increased reports of missing work, loss in productivity, and time spent off work compared to men without arthritis (Tables 5 and 6). These results suggest that men with arthritis are still being affected by the disease from a functional standpoint. They may just be choosing to ignore injuries, which consequently has led to an increase in missing work compared to women (Tables 5 and 6). As mentioned previously, a personal dilemma is presented for individuals with arthritis, however, the dilemma may be even more difficult for men due to traditional gender norms. Stergiou-Kita and colleagues note that heroism, physical strength, toughness, and stoicism are celebrated and can lead to the normalization of accepting risk and enduring physical pain and injuries without complaints.[35]

Combatting arthritis

Arthritis can dramatically affect an individual's life, and it persists into old age.[19] As such, there is a need for individuals to receive appropriate accommodations to assist their daily functioning. The burden of arthritis is increasing with an aging workforce, highlighting the need for evidence-based interventions such as physical activity.[36] However, arthritis-specific barriers may cause physical activity to be difficult, thus different approaches or workarounds should be employed.[36] Furthermore, as previously mentioned, individuals with arthritis face a personal dilemma: do they risk injury to reduce financial burden, or are the effects of the disease so strong that they require time off work?

As mentioned previously, individuals who have a lack of accommodating policies related to sickness absence policies are more likely to engage in sickness

presenteeism[33] and the lack of usage of accommodations is also related to their own perceived need.[34] More inclusive policies are required to demonstrate that it is acceptable to use these policies while also appropriately supporting the individual's needs. Doing so can improve the management of the disease and retain function in an occupational setting. Additionally, self-management strategies such as specific interactive disease education, problem solving, etc. have beneficial effects for the psychosocial impacts on individuals with arthritis. These solutions are also cost-effective, thus garnering inclusive access to individuals with the chronic disease regardless of socioeconomic income.[37] Perhaps the middle ground to the personal dilemma is to increase their cautiousness regarding potential injury at work to avoid the financial dilemma while also reducing their chance of injury. Alternatively, increased awareness can potentially be a caveat. Individuals may associate actual injuries as minor or major flare-ups in their arthritis, especially when they may not be aware of the full extent of their disease, leading to worse actions in response to the disease. Badley and colleagues found that individuals who don't know what type of arthritis they have are likely to have lower health literacy, less access to heathcare and other support services, while also having low income, low education, and of a non-white cultural background.[38] In turn, the intersectional background of a person should be carefully considered when creating better management strategies for the disease. Future research should investigate strategies tailored to different backgrounds in greater depth.

Limitations

There are several limitations within the study, one of the largest being that the data was entirely self-reported. Thus, some survey responses could have been skewed due to participants defining specific terms differently compared to others. Furthermore, variables such as injury are not always reported, so the actual effect size may be larger than initially discovered within the study. Additionally, the sample sizes were uneven across individuals with arthritis and those without. The sample sizes were also uneven across both datasets (i.e., 2015–2016 and 2017–2018) thus any inferences about the increase/decrease in the odds ratio for specific variables must be approached with caution (e.g., individuals less likely to miss work due to a chronic condition in the last three months (Table 5)). A significant limitation is the inability to confirm whether all participants from the 2015 survey joined the second survey or if the same participants were involved. This uncertainty raises concerns about the continuity of the sample and may impact the validity of our analysis. Given these limitations, including potential biases from self-reporting and uneven sample sizes, the generalizability of our findings to broader populations is limited, underscoring the need for further research to validate our conclusions. Additionally, other confounding factors, such as exercise habits, living environment, diet, and rates of obesity, were not addressed in this study. These factors could significantly influence the outcomes and should be considered in future research to provide a more comprehensive understanding of arthritis. Some variables were also unable to be observed in the 2017–2018 dataset due to none of the variables having a significantly strong correlation with any of the other variables within the study (i.e., reason for why an individual did not work at a job/business for a specified reason within the last 3 months and why an individual missed work due to a chronic condition within the last 3 months) and lacked any correlations of a very high strength ($r < .9$). It is also important to note that since there is a lack of confirmation whether the datasets included the same sample (unlikely since they are unequal), it is difficult to go further with statistical tests such as bootstrapping to fully confirm if the composite variables created were accurate depictions of the variables they replaced. Additionally, since this study involves secondary data analysis, we are constrained by how the dataset defines and categorizes variables. This limitation affects our ability to modify or redefine variables, meaning we must work within the framework provided by the existing data.

Conclusion

The study aimed to investigate if individuals with arthritis are more likely to miss work, face injuries and activity limitations, and have difficulty functioning in their daily lives compared to those without a chronic illness. Individuals with arthritis were found to be more likely to miss work, face activity limitations, and have difficulty functioning throughout their daily lives than those without arthritis but have a reduced chance of reporting injury compared to their non-arthritis counterparts. Women with arthritis were found to be more likely to report difficulty with all ADLs and not having a job compared to their male counterparts. Alternatively, men with arthritis were found to be more likely to report work absences and were less likely to report injury compared to their female counterparts. While drawing upon the PEO, the physical effects that can be seen in an individual's life to the psychosocial effects that it has on an individual, the effects of arthritis are far reaching and can have long lasting effects on the person's overall functioning and occupational performance.

Individuals with arthritis are prone to worsening of their symptoms over time[19] and a variety of psychosocial effects that can impact their personal life and their own daily functioning.[12] The impact of the disease can massively affect an individual's work life, leading to increased absenteeism and reduced productivity, which can result ino a sense of learned helplessness.[21] The increased chance of reporting injury also suggests that individuals are also more wary of

their own condition.[21] In turn, future studies should look at the various accommodations that companies currently have, how employees feel about them and why these measures are not being fully utilized despite functionality being clearly affected in individuals with arthritis. The study's results in tandem with current research suggest a gap in accessibility, effectiveness or implementation indicating the need to consider a wider scope of factors such as the functional limitations posed by arthritis. Additionally, the psychosocial effects such as learned helplessness[38] in individuals with arthritis should be investigated further. It is also unclear in the present study whether the psychosocial effects experienced are a matter of the deterioration of arthritis symptoms over time, or if they are due to a matter of age. By differentiating between the two, future solutions can be adopted to focus more on the long-term effects of aging to alleviate/reduce arthritis (e.g., increased social support), or the focus on the physical functioning of individuals as they age.

The study also has functional implications, individuals will need access to a continuous stream of support throughout their lives as the disease deteriorates. Thus, the call for increased social support for aging arthritis patients such as caregivers (e.g., family/friends or hired) who can recognize and implement practices that are more arthritis friendly. Furthermore, women experience greater functional impairments than men, highlighting the disproportionate burden of the disease on women, targeted interventions that uniquely consider women should be accounted for. Alternatively, the workplace needs adjustments from a culture standpoint to encourage the discussion of injuries, doing so can accommodate the lack of reporting injuries prominent in men. Giving additional support for individuals with arthritis also has several clinical implications, therapists will need to watch out for long term effects when it comes to taking care of patients and preparing them for long term as it is a deteriorating disease. Since a large bulk of individuals are in their working years, therapists will also need to implement a holistic approach in guiding patients with arthritis. More specifically, the recognition of more ergonomic practices to further prevent injury and reduce the chances of physical flareups. Additionally, the psychosocial effects that are accompanied by arthritis draws the need for individuals to use self-management strategies to alleviate such effects.

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